CDesk Test Code

You can run the following test code in pei2 server.

0. Tools

```
CDesk tools download -i /mnt/linzejie/CDesk_test/result/0.Tools/test.txt \
-o . -k /mnt3/linzejie/miniconda3/envs/CDesk/etc/asperaweb_id_dsa.openssh -t 5
```

1. bulkRNA

1. 1 preprocess

```
CDesk bulkRNA preprocess \
-i /mnt/linzejie/CDesk_test/data/1.RNA/1.preprocess/test.csv \
-o ./test -s mm10 -t 100 (-bw -te)
```

1. 2 correlation

```
# no batch
CDesk bulkRNA correlation \
-i /mnt/linzejie/CDesk_test/data/1.RNA/2.QC/fpkm.csv \
-o ./no --group /mnt/linzejie/CDesk_test/data/1.RNA/2.QC/group.csv --batch no
# combat
CDesk bulkRNA correlation -i /mnt/linzejie/CDesk_test/data/1.RNA/2.QC/fpkm.csv \
-o ./combat --group /mnt/linzejie/CDesk_test/data/1.RNA/2.QC/group.csv --batch combat
# removeBatchEffect
CDesk bulkRNA correlation -i /mnt/linzejie/CDesk_test/data/1.RNA/2.QC/fpkm.csv \
-o ./removeBatchEffect --group /mnt/linzejie/CDesk_test/data/1.RNA/2.QC/
```

```
group.csv \
--batch removeBatchEffect
```

1. 3 DEG

```
# Deseq2
CDesk bulkRNA DEG deseq2 \
-i /mnt/linzejie/CDesk_test/data/1.RNA/3.DEGD/merged_count.csv \
-o ./deseq2_666 --group /mnt/linzejie/CDesk_test/data/1.RNA/3.DEGD/met
a.csv \
-p --gene /mnt/linzejie/CDesk_test/data/1.RNA/3.DEGD/gene1.txt
# Adjusted t
CDesk bulkRNA DEG adjusted_t \
-i /mnt/linzejie/work/vessel_20250627/downstream/DEG/gfold/fpkm.csv \
-o ./adjust_t --group /mnt/linzejie/CDesk_test/data/1.RNA/3.DEGD/meta_adj
ust.csv \
-p --fc 1 --pval 0.05 --top_num 1000
# Gfold
CDesk bulkRNA DEG gfold \
-i /mnt3/linzejie/20251107/2.mapping/TibetOrgan/marker/deg1.csv \
-o ./gfold1 -t 50 -s pigT2T -p \
--gene /mnt3/linzejie/20251107/2.mapping/TibetOrgan/marker/genes.txt
```

1. 4 enrich

```
# GO + KEGG single sample
CDesk bulkRNA enrich analyze \
-i /mnt/linzejie/CDesk_test/data/1.RNA/4.enrichment/gene_list.txt \
-s mouse -o ./enrich_single --type single
# GO + KEGG multiple samples
CDesk bulkRNA enrich analyze \
-i /mnt/linzejie/CDesk_test/data/1.RNA/4.enrichment/multi.csv \
-o ./enrich_multi -s mouse --type multi

# Custom single sample
CDesk bulkRNA enrich analyze \
-i /mnt/linzejie/CDesk_test/data/1.RNA/4.enrichment/custom.txt \
```

```
--custom /mnt/linzejie/CDesk_test/data/1.RNA/4.enrichment/custom_ref.txt
\
-o ./custom_enrich_single --type single
# Custom multiple samples
CDesk bulkRNA enrich analyze \
-i /mnt/linzejie/CDesk_test/data/1.RNA/4.enrichment/multi1.csv \
--custom /mnt/linzejie/CDesk_test/data/1.RNA/4.enrichment/custom_ref.txt
\
-o ./custom_enrich_multi --type multi

# GSEA
CDesk bulkRNA enrich GSEA \
-i /mnt/linzejie/CDesk_test/data/1.RNA/4.enrichment/gsea_test.csv -o ./gse
a \
-s pig --cols gene_name,log2FC \
(--path /mnt/linzejie/CDesk_test/data/1.RNA/4.enrichment/paths.txt)
```

1. 5 similarity

```
CDesk bulkRNA similarity \
-i /mnt/linzejie/CDesk_test/data/1.RNA/5.Similarity/files1.txt \
--group /mnt/linzejie/CDesk_test/data/1.RNA/5.Similarity/group.csv \
-o ./similarity --batch removeBatchEffect \
(--gene /mnt/linzejie/CDesk_test/data/1.RNA/5.Similarity/Pluripotent_genes
_20.txt
```

1. 6 cluster

```
# Kmean
CDesk bulkRNA cluster kmean \
-i /mnt/linzejie/CDesk_test/data/1.RNA/6.Clustering/merged_fpkm.csv \
-o ./kmeans \
--group /mnt/linzejie/CDesk_test/data/1.RNA/6.Clustering/meta5.csv \
(--gene /mnt/linzejie/CDesk_test/data/1.RNA/6.Clustering/kmean_test.txt)

# WGCNA
CDesk bulkRNA cluster WGCNA \
```

```
-i /mnt/linzejie/CDesk_test/data/1.RNA/6.Clustering/test.csv \
--pheno /mnt/linzejie/CDesk_test/data/1.RNA/6.Clustering/pheno.csv \
--trait tumor -o ./wgcna
```

1. 7 splice

```
# detect
CDesk bulkRNA splice detect \
--b1 /mnt/linzejie/CDesk_test/data/1.RNA/7.splice/b1.txt \
--b2 /mnt/linzejie/CDesk_test/data/1.RNA/7.splice/b2.txt \
--species rmats_test -o ./detect1

# draw
CDesk bulkRNA splice draw \
--b1 /mnt/linzejie/CDesk_test/data/1.RNA/7.splice/draw_b1.txt \
--b2 /mnt/linzejie/CDesk_test/data/1.RNA/7.splice/draw_b2.txt \
--region chr16:+:9000:25000 --species rmats_test -o ./draw1 \
--group /mnt/linzejie/CDesk_test/data/1.RNA/7.splice/grouping.gf \
--l1 sample1 --l2 sample2
```

2. scRNA

2. 1 preprocess

```
# Cellranger
CDesk scRNA preprocess cellranger \
-i /mnt/linzejie/CDesk_test/data/2.scRNA/1.preprocess/cellranger.csv \
-o . --species hg38 -t 100

# Singleron
CDesk scRNA preprocess singleron -i /mnt/linzejie/work/20260105_singler
on/test.csv \
-o . --species galGal6 -t 100

# Dnbc
CDesk scRNA preprocess dnbc \
-i /mnt/linzejie/CDesk_test/data/2.scRNA/1.preprocess/dnbc.csv \
```

```
-o . --species susScr11 -t 100

# DrSeq
CDesk scRNA preprocess drseq \
-i /mnt/linzejie/CDesk_test/data/2.scRNA/1.preprocess/DrSeq.csv \
-o . --species hg38 -t 100
```

2. 2 clustering

```
# scanpy
CDesk scRNA cluster -i /mnt/linzejie/CDesk_test/data/2.scRNA/2.clusterin
g/pbmc.h5 \
-o . -n test --mode scanpy
# seurat
CDesk scRNA cluster -i /mnt/linzejie/CDesk_test/data/2.scRNA/2.clusterin
g/hg19 \
-o . -n test --mode seurat
```

2. 3 annotation

```
# marker
CDesk scRNA annotation marker \
-i /mnt/linzejie/CDesk_test/data/2.scRNA/3.annotation/qry_data.rds \
--marker marker_info.csv --meta umap_clusters -o ./marker

# transfer
CDesk scRNA annotation transfer \
--query /mnt/linzejie/CDesk_test/data/2.scRNA/3.annotation/qry_data.rds \
--ref /mnt/linzejie/CDesk_test/data/2.scRNA/3.annotation/ref_data.rds \
--meta celltype_annotation_20210517 -o ./transfer
```

2. 4 marker

```
# all
CDesk scRNA marker all \
-i /mnt/linzejie/CDesk_test/data/2.scRNA/3.annotation/qry_data.rds \
--marker marker_info.csv --meta umap_clusters -o ./marker
```

```
# 2 group
CDesk scRNA marker 2group \
-i /mnt/linzejie/CDesk_test/data/2.scRNA/4.marker/D1_EMT.filt.rds \
-o ./2group --meta umap_clusters --type1 '3' --type2 '7' -f 0.5 -p 0.05 -m
0.25
```

2. 5 trajectory

```
# Velocity
CDesk scRNA trajectory velocity \
--cellranger /mnt/linzejie/CDesk_test/data/2.scRNA/5.trajectory/RNAvelocit
y/pbmc \
--rds /mnt/linzejie/CDesk_test/data/2.scRNA/5.trajectory/RNAvelocity/test.r
ds \
--genes /mnt/linzejie/CDesk_test/data/2.scRNA/5.trajectory/RNAvelocity/g
ene.txt \
-o ./velocity --meta celltype
# Velocity (no loom, have to specify the species)
CDesk scRNA trajectory velocity \
--cellranger /mnt/linzejie/CDesk_test/data/2.scRNA/5.trajectory/RNAvelocit
y/pbmc1 \
--rds /mnt/linzejie/CDesk_test/data/2.scRNA/5.trajectory/RNAvelocity/test.r
ds \
--genes /mnt/linzejie/CDesk_test/data/2.scRNA/5.trajectory/RNAvelocity/g
ene.txt \
-o ./velocity_noloom --meta celltype \
--gtf /mnt/linzejie/CDesk_test/data/2.scRNA/5.trajectory/RNAvelocity/gene
s.gtf

# Monocle3
CDesk scRNA trajectory monocle \
-i /mnt/linzejie/CDesk_test/data/2.scRNA/5.trajectory/monocle/monocle_te
st.rds \
-m celltype --start NK -o ./monocle

# Diffusion
CDesk scRNA trajectory diffusion \
```

```
-i /mnt/linzejie/CDesk_test/data/2.scRNA/5.trajectory/DiffusionMap/monocle_test.rds \
-o ./diffusion -m celltype --start NK

# CStreet
CDesk scRNA trajectory cstreet \
-i /mnt/linzejie/CDesk_test/data/2.scRNA/5.trajectory/CStreet/ExpressionMatrix_list.txt \
-s /mnt/linzejie/CDesk_test/data/2.scRNA/5.trajectory/CStreet/CellStates_list.txt \
-n test -o ./cstreet
```

2. 6 similarity

```
CDesk scRNA similarity \
--sample1 /mnt/linzejie/CDesk_test/data/2.scRNA/6.similarity/seurat_obj1.rds \
--sample2 /mnt/linzejie/CDesk_test/data/2.scRNA/6.similarity/seurat_obj2.rds \
--meta1 seurat1_clusters --meta2 seurat2_clusters \
--type1 /mnt/linzejie/CDesk_test/data/2.scRNA/6.similarity/type1.txt \
--type2 /mnt/linzejie/CDesk_test/data/2.scRNA/6.similarity/type2.txt \
-o . --nfeatures_used 2000 --integrate_features 2000 --width 8 --height 8
```

2. 7 Interaction

```
# cellchat (long time, > 1h)
CDesk scRNA interaction cellchat \
-i /mnt/linzejie/CDesk_test/data/2.scRNA/7.interaction/cellchat.rds \
-o ./cellchat -t 100 --species human --group celltype --width 8 --height 8

# pyscenic
CDesk scRNA interaction pyscenic \
-i /mnt/yutiancheng/CDesk/CDesk/2.scRNAseq/GRN/data/pbmc3k_preprocessed.h5ad \
--tf /mnt/yutiancheng/CDesk/CDesk/2.scRNAseq/GRN/motifs/human_tfs_test.txt \
```

```
--db /mnt/yutiancheng/CDesk/CDesk/2.scRNAseq/GRN/motifs -t 100 --spe
cies hg38 \
--auc 0.05 --nes 3.0 --normalize False --scale False --log False \
-o ./pyscenic --top_regulons 20 --top_targets 20 (--plot_tf BCL6)
```

2. 8 Integrate

```
# bulk + scRNA
CDesk scRNA integrate bulk \
--bulk /mnt/linzejie/CDesk_test/data/2.scRNA/8.integrate/bulk_data_list.txt
\
--scRNA /mnt/linzejie/CDesk_test/data/2.scRNA/8.integrate/scRNA_data_li
st.txt \
--meta /mnt/linzejie/CDesk_test/data/2.scRNA/8.integrate/meta.csv -o ./bu
lk_scRNA

# Integrate
## CCA
CDesk scRNA integrate seurat \
-i /mnt/linzejie/CDesk_test/data/2.scRNA/8.integrate/merge.csv --mode C
CA -o ./CCA
## harmony
CDesk scRNA integrate seurat \
-i /mnt/linzejie/CDesk_test/data/2.scRNA/8.integrate/merge.csv \
--mode harmony -o ./harmony
## merge
CDesk scRNA integrate seurat \
-i /mnt/linzejie/CDesk_test/data/2.scRNA/8.integrate/merge.csv --mode m
erge -o ./merge
```

3. ATAC

3. 1 preprocess

```
CDesk ATAC preprocess \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/1.preprocess/test.csv \
```

```
-o ./test -t 50 -s mm10
```

3. 2 QC

```
CDesk ATAC QC \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/2.QC/meta.csv \
--bin 100000 -t 100 --species mm10 -o ./QC
```

3. 3 dynamic

```
CDesk ATAC dynamic \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/3.dynamic/group.csv \
-o ./dynamic --species mm10 --remove_cluster CO9,OC1
```

3. 4 motif

```
# homer
## peak
CDesk ATAC motif homer \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/4.motif/Find_homer/homer.csv \
-o ./homer_peak -s mm10 --mode peak
## gene
CDesk ATAC motif homer \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/4.motif/Find_homer/homer_gene.c
sv \
-o ./homer_gene -s mm10 --mode gene

# meme
CDesk ATAC motif meme \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/4.motif/Find_meme/test.csv \
-db /mnt/linzejie/CDesk_test/data/3.ATAC/4.motif/Find_meme/db.txt \
-o ./meme

# homer_heatmap
CDesk ATAC motif homer_heatmap \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/4.motif/homer_heatmap/motif_gro
up.csv \
```

```
--motif /mnt/linzejie/CDesk_test/data/3.ATAC/4.motif/homer_heatmap/p53_p63.txt \
-o ./homer_heatmap

# enrich_heatmap
CDesk ATAC motif enrich_heatmap \
--motif /mnt/linzejie/CDesk_test/data/3.ATAC/4.motif/enrich_heatmap/motif.csv \
--bed /mnt/linzejie/CDesk_test/data/3.ATAC/4.motif/enrich_heatmap/bed.csv \
-o ./enrich_heatmap -t 50 -s mm10
```

3. 5 accessbility

```
# gene mode
## whole
### sample
CDesk ATAC accessbility \
--bw /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/bw.txt \
-o ./gene_whole_sample \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/gene_list.txt \
--mode gene --species mm10 --center whole --method sample
### region_list
CDesk ATAC accessbility \
--bw /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/bw.txt \
-o ./gene_whole_region \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/gene_list.txt \
--mode gene --species mm10 --center whole --method region_list \
--region_list /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/gene_list.txt
## TSS
### sample
CDesk ATAC accessbility \
--bw /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/bw.txt \
-o ./gene_TSS_sample \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/gene_list.txt \
--mode gene --species mm10 --center TSS --method sample
### region_list
```

```
CDesk ATAC accessbility \
--bw /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/bw.txt \
-o ./gene_TSS_region \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/gene_list.txt \
--mode gene --species mm10 --center TSS --method region_list \
--region_list /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/gene_list.t
xt
## TTS
### sample
CDesk ATAC accessbility \
--bw /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/bw.txt \
-o ./gene_TTS_sample \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/gene_list.txt \
--mode gene --species mm10 --center TTS --method sample
### region_list
CDesk ATAC accessbility \
--bw /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/bw.txt \
-o ./gene_TTS_region \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/gene_list.txt \
--mode gene --species mm10 --center TTS --method region_list \
--region_list /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/gene_list.t
xt

# region mode
## position
### sample
CDesk ATAC accessbility \
--bw /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/bw.txt \
-o ./region_position_sample \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/regions_fixed.bed \
--mode region --method sample --center position
### region_list
CDesk ATAC accessbility \
--bw /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/bw.txt \
-o ./region_position_region \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/regions_fixed.bed \
--mode region --method region_list --center position \
--region_list /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/test.txt
```

```
## peak
### sample
CDesk ATAC accessbility \
--bw /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/bw.txt \
-o ./region_peak_sample \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/regions_fixed.bed \
--mode region --method sample --center peak \
--region_list /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/test.txt
### region_list
CDesk ATAC accessbility \
--bw /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/bw.txt \
-o ./region_peak_region \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/regions_fixed.bed \
--mode region --method region_list --center peak \
--region_list /mnt/linzejie/CDesk_test/data/3.ATAC/5.accessbility/test.txt
```

3.6 igv

```
# Sometimes can not load online genome due to network fluctuation
CDesk ATAC igv \
-i /mnt/linzejie/CDesk_test/data/3.ATAC/6.igv/input.txt \
--region /mnt/linzejie/CDesk_test/data/3.ATAC/6.igv/region.txt \
-o ./igv --genome hg19
```

3.7 correlation

```
CDesk ATAC correlation \
--bw1 /mnt/linzejie/CDesk_test/data/3.ATAC/7.Correlation/n70_day0.bw \
--bw2 /mnt/linzejie/CDesk_test/data/3.ATAC/7.Correlation/wt_day0.bw \
--region /mnt/linzejie/CDesk_test/data/3.ATAC/7.Correlation/mm10.2kprom
otoer.bed \
-o ./correlation
```

4. ChIPseq&CUTTag

4. 1 preprocess

```
CDesk ChIPseqCUTTag preprocess \
-i /mnt/linzejie/CDesk_test/data/4.ChIPseqCUTTag/1.preprocess\test.csv \
-o ./test -t 100 -s mm10
```

4. 3 pattern

```
CDesk ChIPseqCUTTag pattern \
-bed /mnt/linzejie/CDesk_test/data/4.ChIPseqCUTTag/3.pattern/bed.txt \
-bw /mnt/linzejie/CDesk_test/data/4.ChIPseqCUTTag/3.pattern/bw.txt -o ./t
est
```

5. HiC

5. 1 preprocess

```
# mm10
CDesk HiC preprocess -i /mnt/linzejie/CDesk_test/data/5.HiC/1.preprocess/
test.csv \
-s mm10 -o ./mm10 -t 100
# mm10_plusmajsat
CDesk HiC preprocess -i /mnt/linzejie/CDesk_test/data/5.HiC/1.preprocess/
test.csv \
-s mm10_plusmajsat -o ./mm10_plusmajsat  \
-t 100 -f -r 100000,1000000
# done
```

5. 2 Correlation

```
# chr15
CDesk HiC correlation -i /mnt/linzejie/CDesk_test/data/5.HiC/2.Correlation/
test.csv \
-o ./chr15 --chr chr15
# All chr
CDesk HiC correlation -i /mnt/linzejie/CDesk_test/data/5.HiC/2.Correlation/
test.csv \
```

```
-o all
# done
```

5. 3 Marix

```
# 1. to fanchic
# 1.1 h5 √
nohup CDesk HiC matrix -i /mnt/linzejie/CDesk_test/data/5.HiC/data/Hi-C_8-cell.100k.h5 -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/h5_to_fanc --inputFormat h5 --outputFormat fanchic --oe 200000 --sparse 200000 -t 50 -n VC -r 200000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/h5_to_fanc.log &
# 1.2 cool √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_8-cell.100k.cool -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/cool_to_fanc --inputFormat cool --outputFormat fanchic --oe 200000 --sparse 200000 -t 50 -n VC -r 200000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/cool_to_fanc.log &
# 1.3 mcool √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_early_2-cell.100000N.mcool -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/mcool_to_fanc --inputFormat mcool --outputFormat fanchic --oe 300000 --sparse 300000 -t 50 -n VC -r 300000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/mcool_to_fanc.log &
# 1.4 juicerhic √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_ICM.hic -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/juicer_to_fanc --inputFormat juicerhic --outputFormat fanchic --oe 300000 --sparse 300000 -t 50 -n KR -r 300000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/juicer_to_fanc.log &
# 1.5 fanchic √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/loop/Hi-C_ICM.100k.ice.hic -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/fanc_to_fanc --inputFormat fanchic --outputFormat fanchic --oe 300000 --sparse 300000 -t 50 -n KR -r 300000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/fanc_to_fanc.log &
# 1.6 pairs √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/tes
```

```
t_matrix/Hi-C_ICM.sort.pairs.gz -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/pairs_to_fanc --inputFormat pairs --outputFormat fanchic --oe 100000 --sparse 100000 -t 50 -n VC -r 100000 --species mm10 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/pairs_to_fanc.log &

# 2. to cool
# 2.1 h5 √
nohup CDesk HiC matrix -i /mnt/linzejie/CDesk_test/data/5.HiC/data/Hi-C_8-cell.100k.h5 -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/h5_to_cool --inputFormat h5 --outputFormat cool --oe 200000 --sparse 200000 -t 50 -n KR -r 200000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/h5_to_cool.log &
# 2.2 cool √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_8-cell.100k.cool -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/cool_to_cool --inputFormat cool --outputFormat cool --oe 200000 --sparse 200000 -t 50 -n VC -r 200000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/cool_to_cool.log &
# 2.3 mcool √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_early_2-cell.100000N.mcool -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/mcool_to_cool --inputFormat mcool --outputFormat cool --oe 300000 --sparse 300000 -t 50 -n VC -r 300000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/mcool_to_cool.log &
# 2.4 juicerhic √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_ICM.hic -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/juicer_to_cool --inputFormat juicerhic --outputFormat cool --oe 300000 --sparse 300000 -t 50 -n KR -r 300000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/juicer_to_cool.log &
# 2.5 fanchic √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/loop/Hi-C_ICM.100k.ice.hic -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/fanc_to_cool --inputFormat fanchic --outputFormat cool --oe 300000 --sparse 300000 -t 50 -n KR -r 300000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/fanc_to_cool.log &
# 2.6 pairs √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/tes
```

```
t_matrix/Hi-C_ICM.sort.pairs.gz -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/pairs_to_cool --inputFormat pairs --outputFormat cool --oe 100000 --sparse 100000 -t 50 -n VC -r 100000 --species mm10 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/pairs_to_cool.log &

# 3. to h5
# 3.1 h5 √
nohup CDesk HiC matrix -i /mnt/linzejie/CDesk_test/data/5.HiC/data/Hi-C_8-cell.100k.h5 -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/h5_to_h5 --inputFormat h5 --outputFormat h5 --oe 200000 --sparse 200000 -t 50 -n KR -r 200000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/h5_to_h5.log &
# 3.2 cool √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_8-cell.100k.cool -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/cool_to_h5 --inputFormat cool --outputFormat h5 --oe 200000 --sparse 200000 -t 50 -n VC -r 200000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/cool_to_h5.log &
# 3.3 mcool √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_early_2-cell.100000N.mcool -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/mcool_to_h5 --inputFormat mcool --outputFormat h5 --oe 300000 --sparse 300000 -t 50 -n VC -r 300000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/mcool_to_h5.log &
# 3.4 juicerhic √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_ICM.hic -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/juicer_to_h5 --inputFormat juicerhic --outputFormat h5 --oe 300000 --sparse 300000 -t 50 -n KR -r 300000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/juicer_to_h5.log &
# 3.5 fanchic √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/loop/Hi-C_ICM.100k.ice.hic -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/fanc_to_h5 --inputFormat fanchic --outputFormat h5 --oe 300000 --sparse 300000 -t 50 -n KR -r 300000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/fanc_to_h5.log &
# 3.6 pairs √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_ICM.sort.pairs.gz -o /mnt/linzejie/CDesk_test/result/5.HiC/m
```

```
atrix/pairs_to_h5 --inputFormat pairs --outputFormat h5 --oe 100000 --spa
rse 100000 -t 50 -n VC -r 100000 --species mm10 > /mnt/linzejie/CDesk_te
st/result/5.HiC/matrix/pairs_to_h5.log &

# 4. to mcool
# 4.1 h5 √
nohup CDesk HiC matrix -i /mnt/linzejie/CDesk_test/data/5.HiC/data/Hi-C_8
-cell.100k.h5 -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/h5_to_mcool -
-inputFormat h5 --outputFormat mcool --oe 200000 --sparse 200000 -t 5
0 -n KR -r 200000,600000 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/h
5_to_mcool.log &
# 4.2 cool √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/tes
t_matrix/Hi-C_8-cell.100k.cool -o /mnt/linzejie/CDesk_test/result/5.HiC/matr
ix/cool_to_mcool --inputFormat cool --outputFormat mcool --oe 300000 --
sparse 300000 -t 50 -n VC -r 300000,600000 > /mnt/linzejie/CDesk_test/r
esult/5.HiC/matrix/cool_to_mcool.log &
# 4.3 mcool √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/tes
t_matrix/Hi-C_early_2-cell.100000N.mcool -o /mnt/linzejie/CDesk_test/resul
t/5.HiC/matrix/mcool_to_mcool --inputFormat mcool --outputFormat mcool
--oe 300000 --sparse 300000 -t 50 -n VC -r 300000,600000 > /mnt/linzej
ie/CDesk_test/result/5.HiC/matrix/mcool_to_mcool.log &
# 4.4 juicerhic √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/tes
t_matrix/Hi-C_ICM.hic -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/juicer
_to_mcool --inputFormat juicerhic --outputFormat mcool --oe 300000 --sp
arse 300000 -t 50 -n KR -r 300000,600000 > /mnt/linzejie/CDesk_test/res
ult/5.HiC/matrix/juicer_to_mcool.log &
# 4.5 fanchic √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/loo
p/Hi-C_ICM.100k.ice.hic -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/fan
c_to_mcool --inputFormat fanchic --outputFormat mcool --oe 300000 --sp
arse 300000 -t 50 -n KR -r 300000,600000 > /mnt/linzejie/CDesk_test/res
ult/5.HiC/matrix/fanc_to_mcool.log &
# 4.6 pairs √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/tes
t_matrix/Hi-C_ICM.sort.pairs.gz -o /mnt/linzejie/CDesk_test/result/5.HiC/m
```

```
atrix/pairs_to_mcool --inputFormat pairs --outputFormat mcool --oe 30000
0 --sparse 300000 -t 50 -n VC -r 300000,600000 --species mm10 > /mnt/
linzejie/CDesk_test/result/5.HiC/matrix/pairs_to_mcool.log &

#  5. to juicerhic
# 5.1 h5 √
nohup CDesk HiC matrix -i /mnt/linzejie/CDesk_test/data/5.HiC/data/Hi-C_8
-cell.100k.h5 -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/h5_to_juicerhi
c --inputFormat h5 --outputFormat juicerhic --oe 300000 --sparse 300000
-t 50 -n KR -r 300000 --species mm10 > /mnt/linzejie/CDesk_test/result/5.
HiC/matrix/h5_to_juicerhic.log &
# 5.2 cool √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/tes
t_matrix/Hi-C_8-cell.100k.cool -o /mnt/linzejie/CDesk_test/result/5.HiC/matr
ix/cool_to_juicerhic --inputFormat cool --outputFormat juicerhic --oe 3000
00 --sparse 300000 -t 50 -n VC -r 300000,600000 --species mm10 > /mn
t/linzejie/CDesk_test/result/5.HiC/matrix/cool_to_juicerhic.log &
# 5.3 mcool √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/tes
t_matrix/Hi-C_early_2-cell.100000N.mcool -o /mnt/linzejie/CDesk_test/resul
t/5.HiC/matrix/mcool_to_juicerhic --inputFormat mcool --outputFormat juic
erhic --oe 300000 --sparse 300000 -t 50 -n VC -r 300000,600000 --spec
ies mm10 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/mcool_to_juicerhi
c.log &
# 5.4 juicerhic √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/tes
t_matrix/Hi-C_ICM.hic -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/juicer
_to_juicerhic --inputFormat juicerhic --outputFormat juicerhic --oe 300000
--sparse 300000 -t 50 -n KR -r 300000 --species mm10 > /mnt/linzejie/CD
esk_test/result/5.HiC/matrix/juicer_to_juicerhic.log &
# 5.5 fanchic √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/loo
p/Hi-C_ICM.100k.ice.hic -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/fan
c_to_juicerhic --inputFormat fanchic --outputFormat juicerhic --oe 300000
--sparse 300000 -t 50 -n KR -r 300000,600000 --species mm10 > /mnt/lin
zejie/CDesk_test/result/5.HiC/matrix/fanc_to_juicerhic.log &
# 5.6 pairs √
nohup CDesk HiC matrix -i /mnt/zhangzheting/cdesk_development/HIC/tes
```

```
t_matrix/Hi-C_ICM.sort.pairs.gz -o /mnt/linzejie/CDesk_test/result/5.HiC/matrix/pairs_to_juicerhic --inputFormat pairs --outputFormat juicerhic --oe 300000 --sparse 300000 -t 50 -n VC -r 300000,600000 --species mm10 > /mnt/linzejie/CDesk_test/result/5.HiC/matrix/pairs_to_juicerhic.log &
```

5. 4 TAD

```
# test1
CDesk HiC TAD \
-i /mnt/linzejie/CDesk_test/data/5.HiC/4.TAD/test1.csv \
-o ./test1 \
--insulation_window 1000000 --directionality_window 1000000 \
--expand 1000000
# test2
CDesk HiC TAD \
-i /mnt/linzejie/CDesk_test/data/5.HiC/2.Correlation/test.csv \
-o ./test2 \
--insulation_window 1000000 --directionality_window 1000000 \
--expand 1000000
# done
```

5. 5 Compartment

```
# with GC
CDesk HiC compartment \
-i /mnt/linzejie/CDesk_test/data/5.HiC/6.compartment/test.csv \
-o ./GC \
--GC /mnt/zhangzheting/pubdata/Xiewei/l2c/wt.100k.atac.bed \
-t 100 --width 10 --height 8 -r 100000 -s mm10
# done

# no GC
CDesk HiC compartment \
-i /mnt/linzejie/CDesk_test/data/5.HiC/6.compartment/test.csv \
-o ./no_GC \
-t 100 --width 10 --height 8 -r 100000 -s mm10
# done
```

5. 6 Loop

```
CDesk HiC loop \
-i /mnt/zhangzheting/cdesk_development/HIC/loop/Hi-C_ICM.100k.ice.hic \
-o ./6.loop -t 50
```

5. 7 3D reconstruction

```
# Flamingo
CDesk HiC reconstruct flamingo \
-i /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_8-cell.hic \
-o ./flamingo --chr chr1 --domain_res 1e6 --frag_res 5e4 --normalization K
R -t 50
# done

# hickit
## no phase
CDesk HiC reconstruct hickit \
--r1 /mnt/zhangzheting/pubdata/Xiewei/allelic/Hi-C_ICM_rep3/Hi-C_ICM_re
p3_1_val_1.fq.gz \
--r2 /mnt/zhangzheting/pubdata/Xiewei/allelic/Hi-C_ICM_rep3/Hi-C_ICM_r
ep3_2_val_2.fq.gz \
--species mm10 --prefix nophase -t 100 -o ./hickit_nophase
## Distance-contact
CDesk HiC reconstruct hickit \
--vcf /mnt/zhangzheting/reference/mm10/mgp.v5.merged.snps_all.dbSNP1
42.vcf.gz \
--mat C57BL_6NJ --pat PWK_PhJ \
--r1 /mnt/zhangzheting/pubdata/Xiewei/allelic/Hi-C_ICM_rep3/Hi-C_ICM_re
p3_1_val_1.fq.gz \
--r2 /mnt/zhangzheting/pubdata/Xiewei/allelic/Hi-C_ICM_rep3/Hi-C_ICM_r
ep3_2_val_2.fq.gz \
--species mm10 --prefix nophase -t 100 -o ./hickit_phase
```

5. 8 Distance contact

```
CDesk HiC pss -i /mnt/linzejie/CDesk_test/data/5.HiC/2.Correlation/test.csv
\
-o ./8.pss --bin 20 --curve_mean_min 1e-6 --curve_bin_min 0 \
--curve_x_min 4 --curve_x_max 8 --expcurve_y_bias -4 --expcurve_x_min
5 \
--expcurve_x_max 8 --heatmap_x_min 4.8 --heatmap_x_max 7.5 --width 10
--height 6
# done
```

5.9 Contact compare

```
CDesk HiC compare \
--hic1 /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_early_2
-cell.100000N.mcool@100000 \
--hic2 /mnt/zhangzheting/cdesk_development/HIC/test_matrix/Hi-C_8-cell.
100k.cool \
-o ./9.compare --chr chr5
# done
```